7. $z = -4x^2 + 2026xy - 2y^2 + 5x$

$$E = \frac{f'(x) \cdot x}{f(x)}$$

$$E_x = \frac{\partial z}{\partial x} \cdot \frac{x}{z} \; ; \quad E_y = \frac{\partial z}{\partial y} \cdot \frac{y}{z}$$

$$\frac{\partial z}{\partial x} = -8x + 2026y + 5 \quad (1;0)$$

$$\frac{\partial z}{\partial x}(1;0) = -8 + 5 = -3$$

$$z(1;0) = -4 + 5 = 1$$

$$E_x = \frac{-3 \cdot 1}{1} = -3$$

$$E_y = \frac{-4 \cdot 1}{-2} = 2$$

$(0;1)$, x, y

$$\frac{\partial z}{\partial y} = -4$$

$$z = -2$$

$z = x^2 - 4xy + 2y^2 + 2x - 4y$ $\cancel{(1; -1)}$

? Будет ли т. экстремума?

$$\int \left(\frac{dx}{\sqrt{x} + \sqrt[3]{x}} \right)$$

$\frac{1}{\sqrt[4]{x}} = x^{-\frac{1}{4}}$

$x^{1/2} = t$

$x^{1/3} = t$?

$\left(\frac{1}{2} x^{-1/2} + \frac{1}{3} x^{-2/3} \right) dx = dt$

$t = x^{1/2} + x^{1/3}$

$t = \sqrt[6]{x} \longrightarrow dt = \frac{1}{6} \frac{dx}{x^{5/6}}$

$6x^{5/6} dt = dx$

$\sqrt{x} = x^{1/2} = t^3$!

$\Rightarrow 6t^5 dt = dx$

$(x^{1/6})^3 = x^{1/2}$

$x^{1/6} = t$

$x^{1/3} = t^2$

$(x^{1/6})^5 = t^5$

$$\int \frac{6t^5 dt}{t^3 + t^2} = \int \frac{6(t^5)}{t^3 + t^2} =$$

!

$t^n = (x^{1/6})^n = x^{1/3} =$

$t^n = x^{1/3}$

$n \cdot \frac{1}{6} \cancel{=} \frac{1}{3} \rightarrow \frac{1}{3} : \frac{1}{6} = \frac{6}{3} = 2$

$n = 2$

$= 6 \int \frac{t^5}{t^3+t^2} dt = 6 \int \frac{t^5}{t^2(t+1)} dt =$

$= 6 \int \frac{t^3}{t+1} dt$

$\frac{t^3 - 1 + 1}{t+1} = \frac{(t+1)(t^2-t+1)-1}{t+1}$

$t^3 + 1 = (t+1)(t^2-t+1)$

$\int \left(t^2 - t + 1 - \frac{1}{t+1} \right) dt =$

$$\int (t^2 - t + 1)\,dt - \int \frac{du}{u} =$$

$$= \frac{t^3}{3} - \frac{t^2}{2} + t - \ln u + C$$

$x^{1/6} = t$

$u = x^{1/6} + 1$

$$\boxed{\frac{\sqrt{x}}{3} - \frac{\sqrt[3]{x}}{2} + \sqrt[6]{x} - \ln(\sqrt[6]{x} + 1) + C}$$

$$\int \frac{1}{x + \sqrt[3]{x}}\,dx$$

$x^{1/3 \cdot 1} = t$

$x^{1/3} = t$

$x^{1/3} = t$

$x^{n/3} = t^n$

$\frac{n}{3} = \frac{2}{3}$

$dt = \frac{1}{3} x^{-2/3} dx$

$dt = \frac{1}{x^{2/3} \cdot 3} dx$

$x^{2/3} = t^2$

$dt = \frac{dx}{3t^2}$

$x^1 = ?$

$x^{\frac{n}{3}} = t^n$

$\frac{n}{3} = 1$

$n = 3$

$3t^2 dt = dx$

$$\int \frac{3t^2 dt}{t^3 + t} =$$

$$= \int \frac{3t^2 dt}{t(t^2+1)} =$$

$$\int \frac{3t\,dt}{t^2+1}$$

$t^2 + 1 = u$

$2t\,dt = du$

$$\int \frac{3\,du}{2(u)} =$$

$$\frac{3}{2} \ln u + C$$

$u = t^2 + 1$

$t = x^{1/3}$

$\rightarrow \frac{3}{2} \ln(x^{2/3} + 1) + C$

$$(\operatorname{tg} x)' = \frac{1}{x^2+1}$$

$$\int \frac{dt}{t^2+1} = \operatorname{tg}(t^2+1)+C$$

$$= \operatorname{tg}(e^x+1)+C$$

$$1-x^2 = t \longrightarrow 1-t = x^2$$

$$-2x\,dx = dt$$

$$dx = \frac{dt}{-2x}$$

$$\int \frac{x^2}{-2}\,\frac{dx}{t^{1/2}} = \int \frac{1-t}{-2t^{1/2}} =$$

$$\int \frac{t-1}{2\sqrt{t}}\,dt$$

$$\sqrt{1-x^2} = t \longrightarrow 1-x^2 = t^2$$

$$\frac{1}{2\sqrt{1-x^2}}\,{-2x}\,dx = dt \qquad 1-t^2 = x^2$$

$$-x^2 = t^2-1$$

$$dx = \frac{\cancel{2}t}{\cancel{2}x}\,dt$$

$$\int \frac{x^{\cancel{2}}\,\cancel{t}\,dt}{-\cancel{x}\cdot\cancel{t}} = \int -x^2\,dt$$

$$= \int (t^2 - 1)\,dt =$$

$$\frac{t^3}{3} - t + C$$

$$\frac{\left(\sqrt{1-x^2}\right)^3}{3} - \sqrt{1-x^2} + C$$

$$\int x^2\sqrt{2x+1}\;dx \qquad dx = t\cdot dt$$

$$\int x^2\,t\cdot t\,dt = \int x^2 t^2\,dt$$

$$\sqrt{2x+1} = t$$

$$2x+1 = t^2$$

$$2x = t^2 - 1$$

$$x = \frac{t^2-1}{2} \mapsto x^2 = \left(\frac{t^2-1}{2}\right)^2$$

$$\int \left(\frac{t^2-1}{2}\right)^2 t^2 \, dt = \int \left(\frac{t^4 - 2t^2 + 1}{4}\right) \cdot t^2 \, dt$$

$$= \int \frac{t^6 - 2t^4 + t^2}{4} \, dt$$

$$\frac{1}{4}\left(\frac{t^7}{7} - \frac{2t^5}{5} + \frac{t^3}{3}\right) + C$$

$$t = \sqrt{2x+1}$$

$$\int x^3 \sqrt{3x+2} \, dx \qquad \frac{3}{2\sqrt{3x+2}} \, dx = dt$$

$dx = \frac{2}{3} t \, dt$

$$\int \frac{2}{3} t^2 x^3 \, dt$$

$$x^3 = \left(\frac{t-2}{3}\right)^3$$

$$\boxed{\int \frac{(t-2)^3}{3^3} t^2 \cdot \frac{2}{3} \, dt}$$